

Digital Logic

CSIT — 1st Semester — 2080 — Answer Sheet
Subject Code: CSC116

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

1. What is combinational circuit? Design a combinational circuit with four inputs lines that represent a decimal digit in BCD and four output lines that generate the 1's complement of the input binary patterns.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. What is asynchronous counter? Design synchronous counter that counts the sequences of 0-1-4-6-7 using T flip-flop.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Implement the Boolean function $F(P,Q,R,S) = \sum (3,4,6,8,9,14)$ using: a. 8 to 1 multiplexer b. PLA c. Decoder

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Attempt any Eight questions [8×5=40]

4. Perform the following operations: a. $(011101)_2 - (110011)_2$ using 2's complement b. $(89344)_{10} - (98654)_{10}$ using 9's complement

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. If $f(P,Q,R,S) = \sum (3,4,7,8,14)$ and $d(P,Q,R,S) = \sum (1,6,9,13)$. Simplify it using K-map and design circuit using minimum number of NAND gates.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. What is drawback of RS Flipflop? Explain D Flip Flop in detail with Logic Diagram, characteristics table and Characteristics equation.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Design a full subtractor with necessary tables and logic diagram.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. What is shift register? Explain 4-bit SISO and PIPO with timing Diagram.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Design an asynchronous Mod 11 up counter using T flip flop.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. How race condition in JK flip flop can be resolved? Explain.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. What is decoder circuit? Design 3 to 8 decoder circuit.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Write short notes on: a. State Diagram b. Encoder c. Parallel Adder

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.